

Question Paper

Month and Year of Examination : **JULY 2026**

Programme : **B.Tech**

Subject: **Analog Electronics**

Course No: **ECPC-209**

Semester: **4th**

Maximum Marks: **50**

Time Allowed: **3 Hrs.**

Total No. of Pages Used: **3**

Number of Questions to be Attempted: **5**

Total No. of Questions: **5**

Unless stated otherwise, the Symbols have their usual meanings in context with the Subject. Assume suitably and state, additional data required, if any.

The Candidates, before starting to write the solutions, should please check the Question Paper for any discrepancy, and also ensure that they have been delivered the question paper of **right course no.** and right subject title.

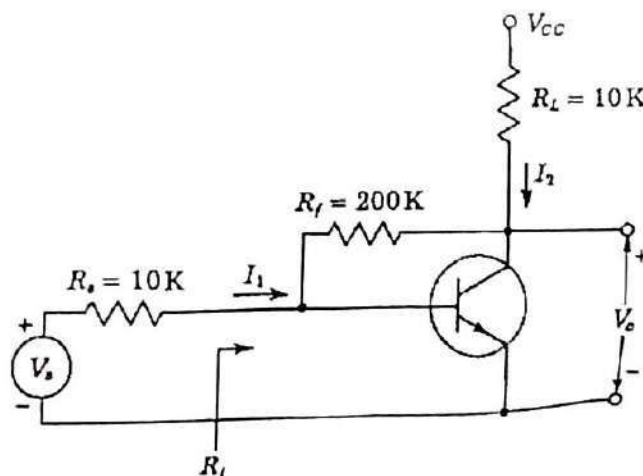
Note: Attempt all the questions with equal marks.

1a. Critically compare the difference in analysis and design of a single stage amplifier based on a MOS and a BJT? Illustrate with the help of suitable mathematics and illustrations.

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1b. Describe in detail why a BJT is modeled using h-parameters. Justify and derive the criterion for simplifying this h-parameter model for a CE configuration.

Find the input impedance R_i in terms of CE h-parameters and R_L in the circuit shown below



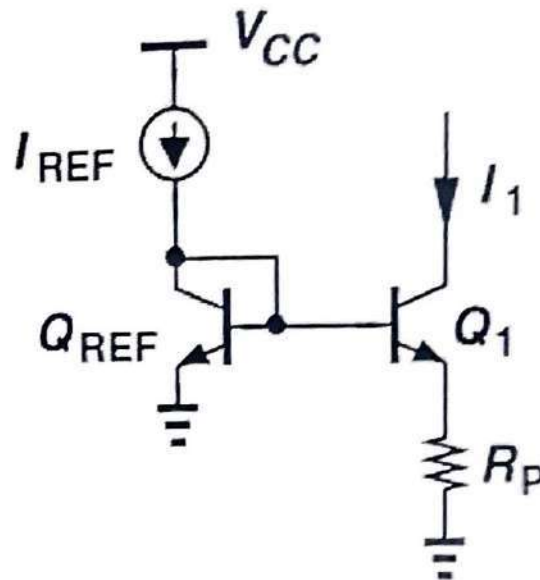
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2a. Classify the amplifiers based on the position of Q point. Also, explain why multiple stages of an amplifier have to be used. Give different coupling methods used in such cascading along with their relative advantages and disadvantages.

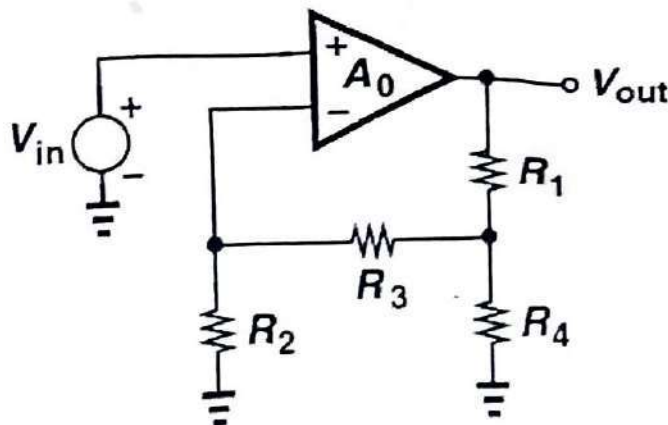
It is desired that the voltage gain of an RC-coupled amplifier at 60 Hz should not decrease by more than 10% from its midband value. Show that the coupling capacitance C must be at least equal to $5.5 / R'$, where $R' = R_o + R_i$ is expressed in K-ohms and C in microfarads.

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- 2b. Define current mirror circuit, its working and utility in differential amplifiers. Due to manufacturing error, resistor R_p has appeared in series with the emitter of Q_1 shown in the Fig. below. Express the value of R_p in terms of other circuit parameters if i) I_1 is half and ii) I_1 is double of its nominal value. Assume Q_{REF} and Q_1 are identical and $\beta \gg 1$

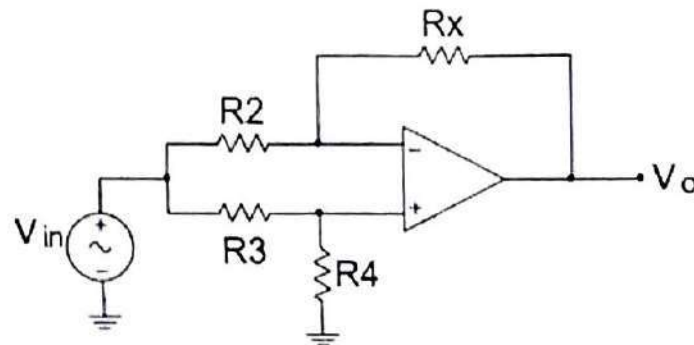


- 3a. Draw and explain equivalent circuit of an OPAMP thus deliberate on its various characteristics as a better amplifier. Calculate the closed loop gain of the non-inverting amplifier as shown in the Fig. if $A_0 = \infty$. Verify that the result reduces to expected values if $R_1 \rightarrow 0$ or $R_3 \rightarrow 0$



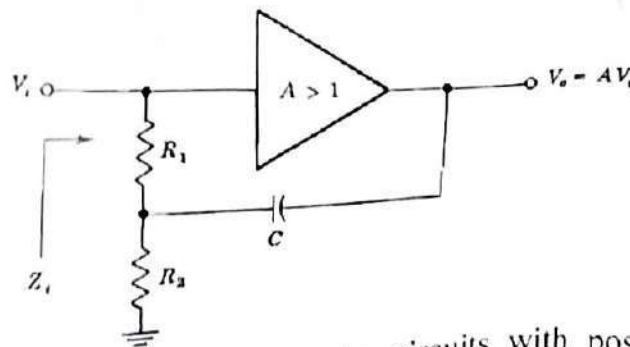
- 3b. An inverting integrator uses an op-amp with input resistor $R = 100 \text{ k}\Omega$ and feedback capacitor $C = 0.1 \mu\text{F}$. The input is a square wave: $+1 \text{ V}$ for 5 ms , then 0 V for 5 ms , repeating. i) Write the expression for the output voltage $v_o(t)$ during the 5 ms interval when $+1 \text{ V}$ is applied ii) Find the output voltage at the end of this 5 ms pulse, assuming the integrator was initially at 0 V iii) Discuss how a parallel resistor across C would change the long-term behaviour.

- 4a. A unity-gain Sallen-Key low-pass filter is required with cutoff frequency $f_c = 1 \text{ kHz}$ and quality factor $Q = 0.707$ (Butterworth). Assume $C_1 = C_2 = 10 \text{ nF}$
- i) Derive the relations between R_1, R_2, C_1, C_2, f_c , and Q for a unity-gain configuration ii) Calculate suitable values of R_1 and R_2 iii) Estimate the magnitude of the gain at $f = 100 \text{ Hz}$ and at $f = 10 \text{ kHz}$.
- 4b. Design an op-amp circuit to perform the following mathematical operation: $V_o = 3V_1 - 5V_2$. Also, for the circuit shown in the Fig. below, determine the value of resistor R_x so that the output voltage is zero.



OR

- 4a. Explain any two techniques of converting analog signal to a digital signal. An 8-bit ADC outputs all 1's when V_i is 5.1V. Find its i) Resolution ii) Digital output when V_i is 1.28V
- 4b. Show that the circuit shown in the Fig. below can simulate a grounded inductor if $A > 1$. In other words, show that the reactive part of Z_i is positive.



5. Discuss the concept of stability in the circuits with positive feedback and subsequently outline and define the conditions for instability in such circuits. Give different types of oscillators with their specific applications. Find V_o/V_i in the circuit shown below and verify that it is not possible to obtain 180° phase shift with a finite attenuation.

